

Problem 29: Unit Step Function

The unit function, also called Heaviside's unit function, $u(t)$ is defined as

$$y = u(t) = 1 \quad t \geq 0$$

$$y = 0 \quad t \leq 0$$

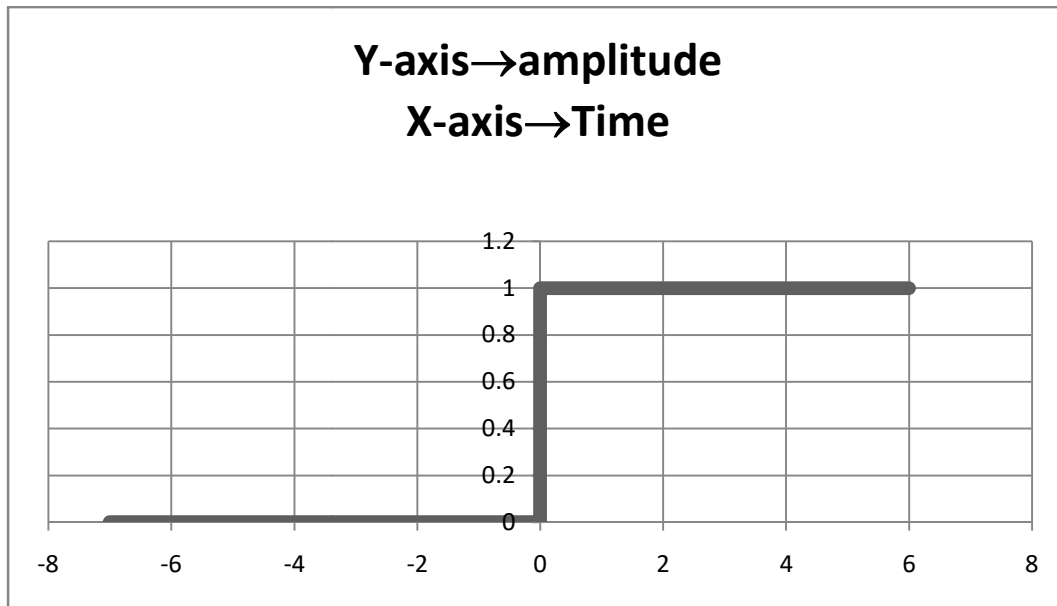


Figure 104

Example 77:

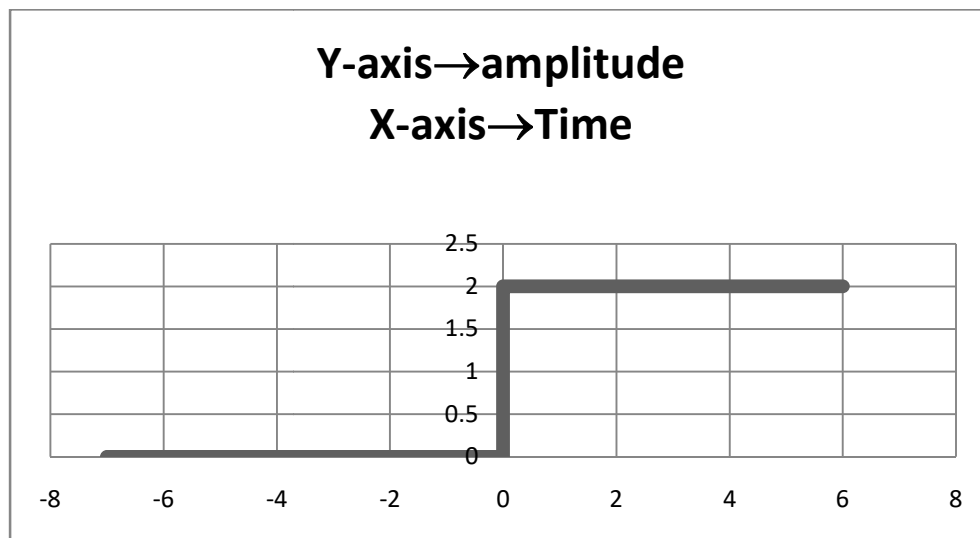


Figure 105

Example 78:

$$y = u(t) = \begin{cases} 1; & t \geq 0 \\ 0; & t < 0 \end{cases}$$

$$\Rightarrow y = u(t-2) = \begin{cases} 1; & t \geq 2 \\ 0; & t < 2 \end{cases}$$

$u(t-2)$

Here,

$$t - 2 = 0$$

$$\therefore t = 2$$

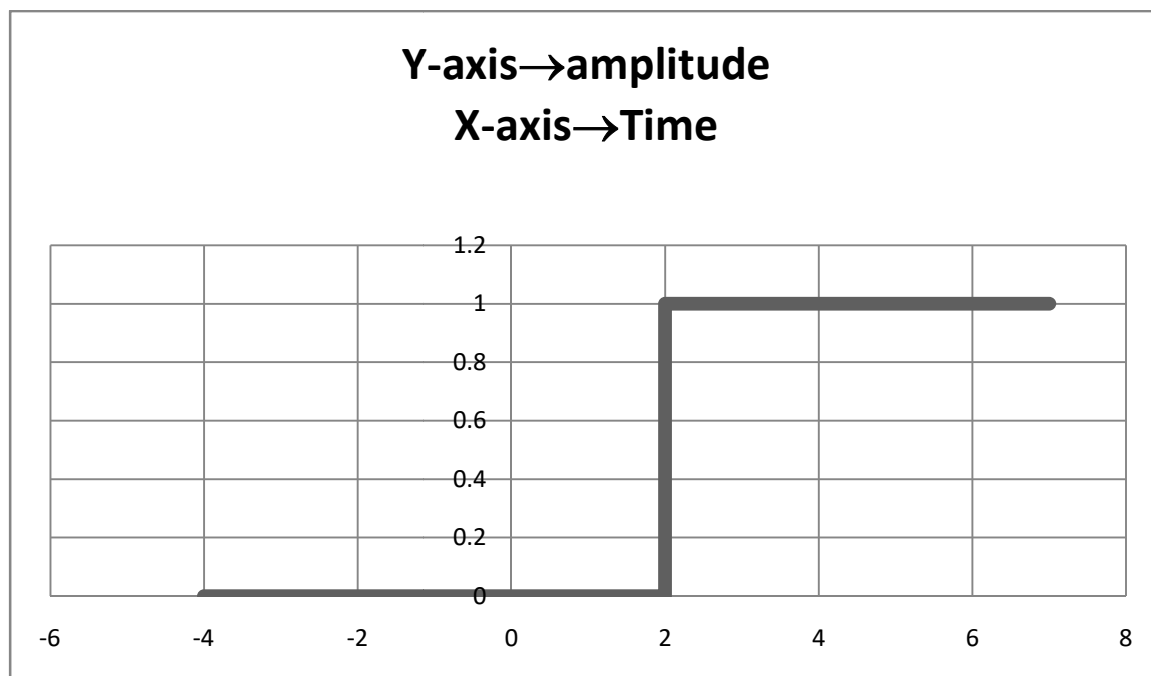


Figure 106

Example 79:

$$y = u(t) = \begin{cases} 1; & t \geq 0 \\ 0; & t < 0 \end{cases}$$

$$\Rightarrow y = u(t-1) = \begin{cases} 1; & t \geq 1 \\ 0; & t < 1 \end{cases}$$

$u(t-1)$

Here,

$$t - 1 = 0$$

$$\therefore t = 1$$

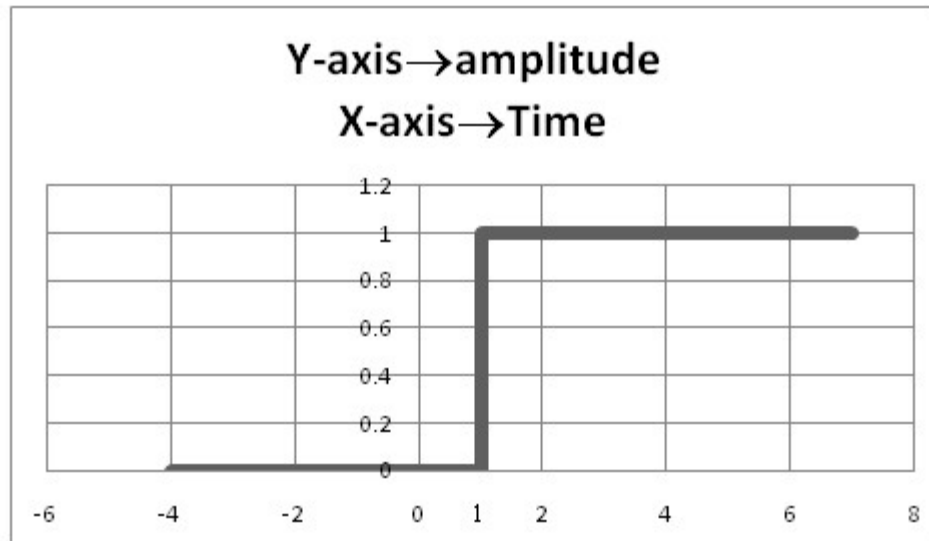


Figure 107

Example 80:

$u(t+1)$

Here,

$$t + 1 = 0$$

$$\therefore t = -1$$

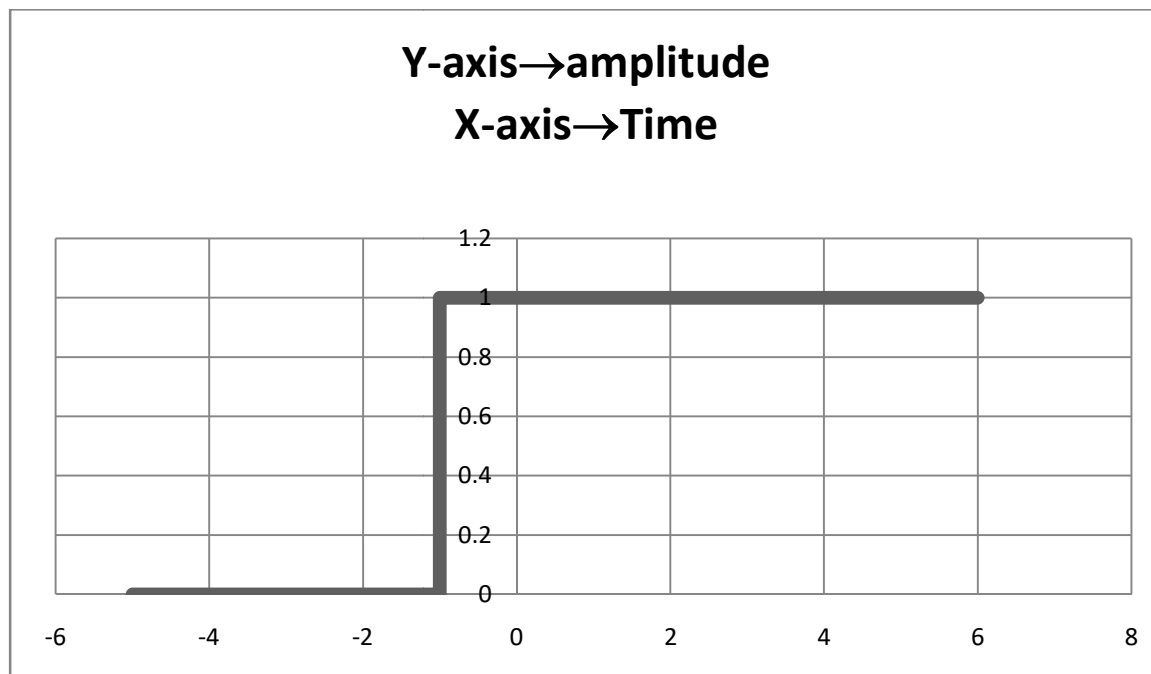


Figure 108

Example 81: $u(t+2)$

Here,

$$t + 2 = 0$$

$$\therefore t = -2$$

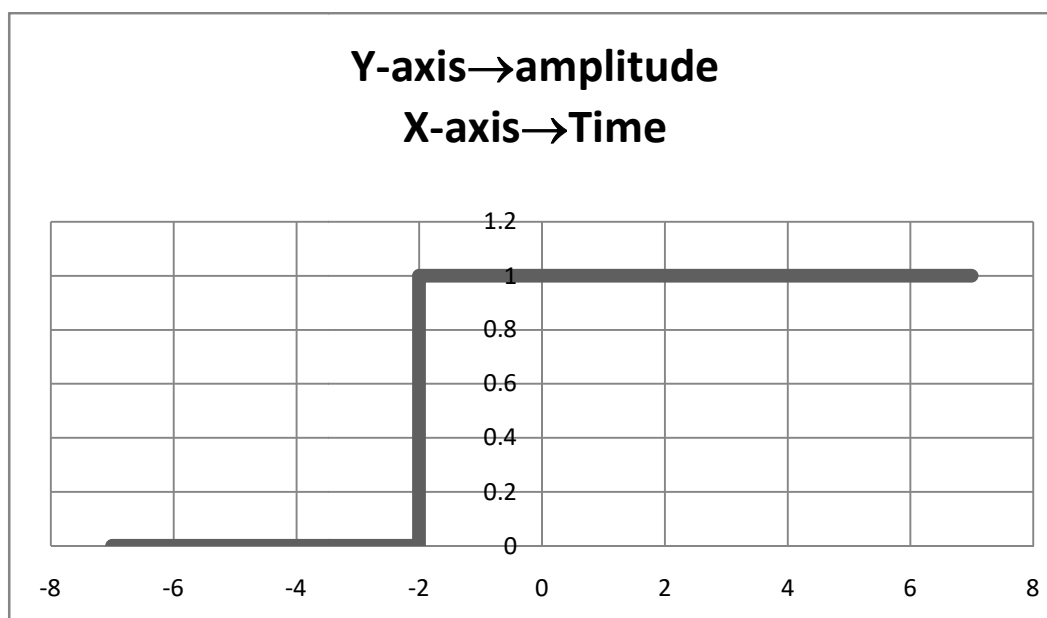


Figure 109

Example 82: $-u(t)$

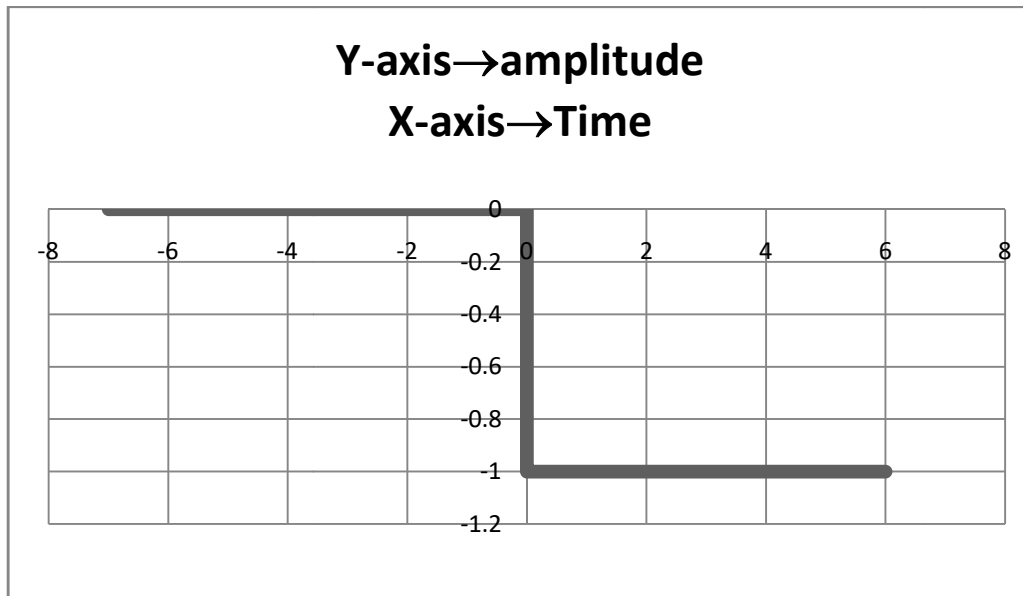


Figure 110

Example 83: $-2u(t)$

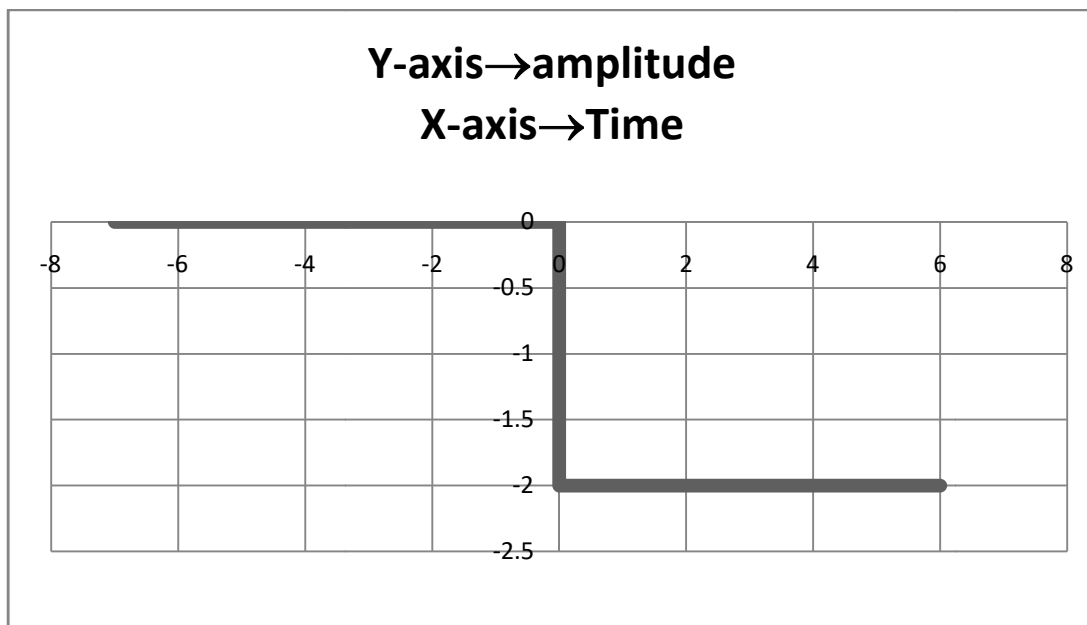


Figure 111

Example 84: $-2u(t-1)$
 Amplitude start from

$$y = -2u(t-1) = -2 ; t \geq 1$$

$$= 0 ; t < 1$$

$$t-1 = 0 \text{ then } t = 1$$

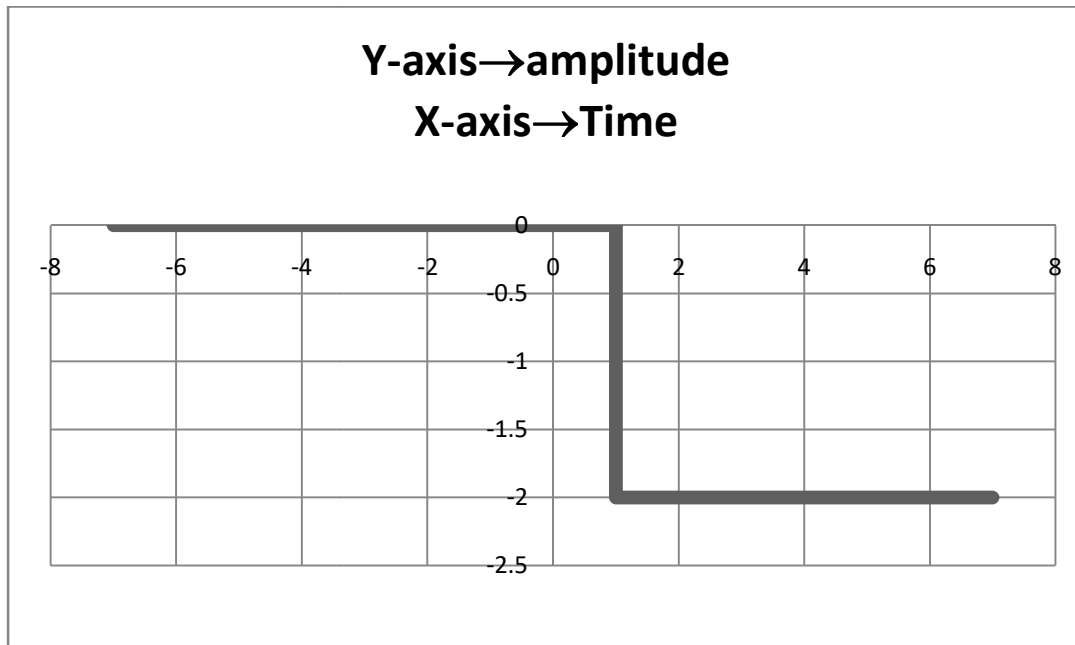


Figure 112

Example 85: $u(t)$

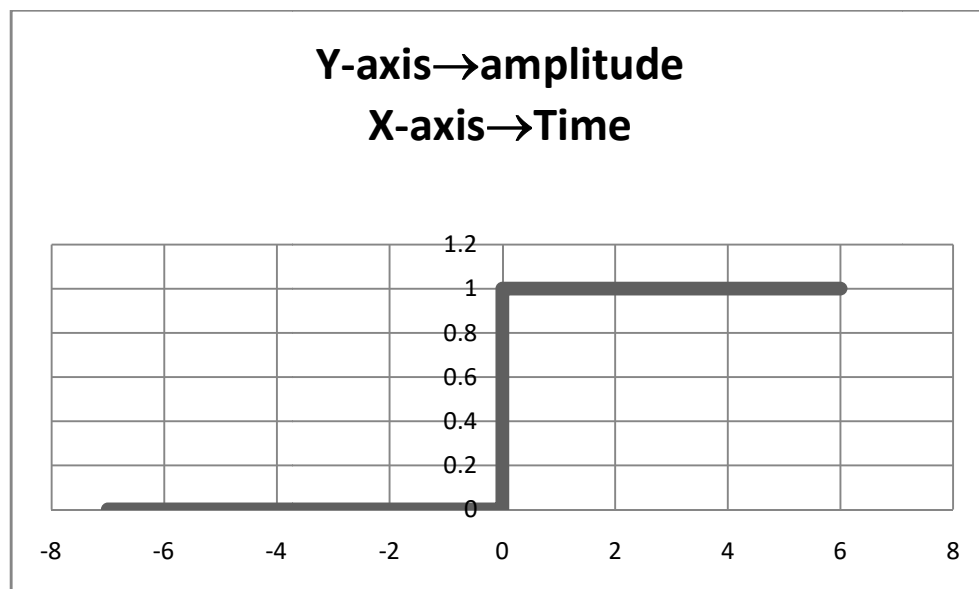


Figure 113

Example 86: $-u(t-2)$

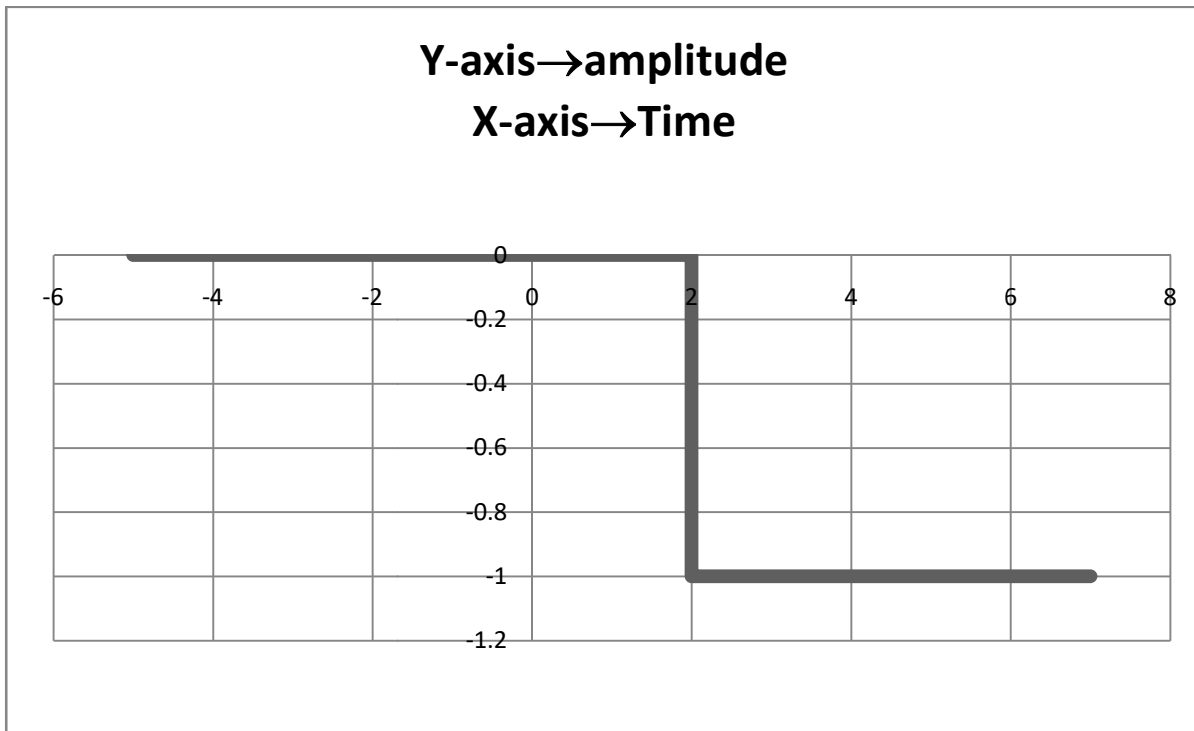


Figure 114

Example 87: $u(t)-u(t-2)$

$$= u(t) + \{-u(t-2)\}$$

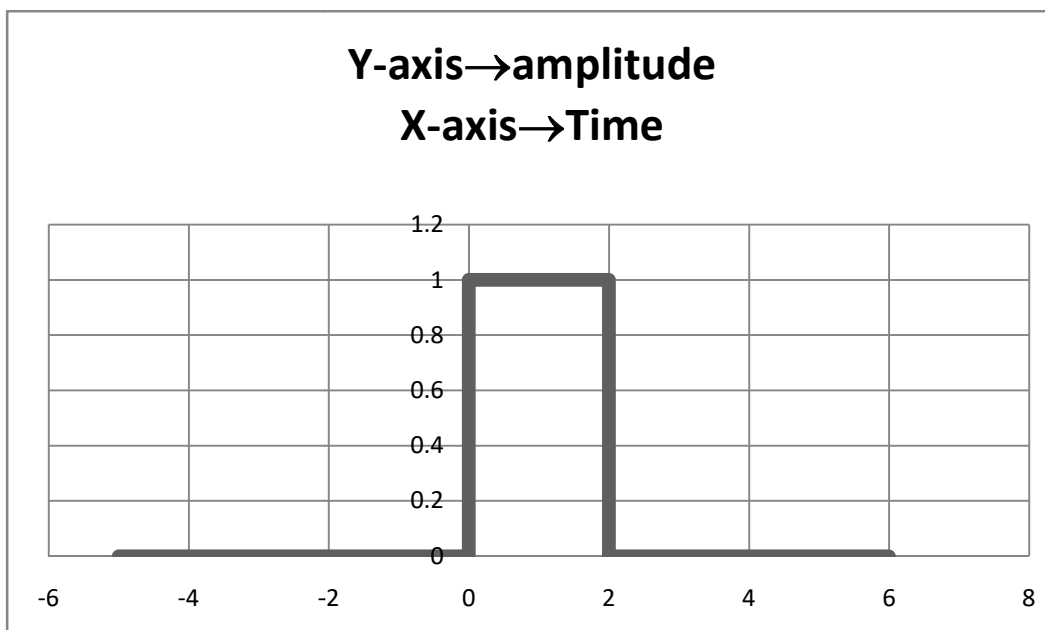


Figure 115

Example 88: $u(t-\pi/2)$

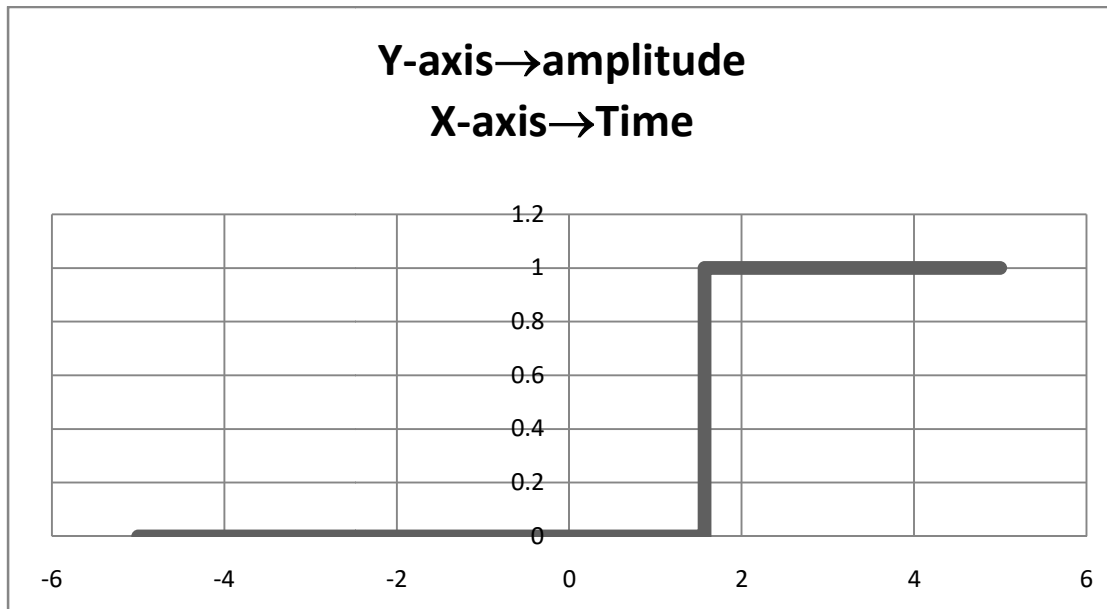


Figure 116

Problem 30: Ramp Function

The ramp function $r(t)$ is defined as

$$y = r(t) = \begin{cases} t & t \geq 0 \\ 0 & t \leq 0 \end{cases}$$

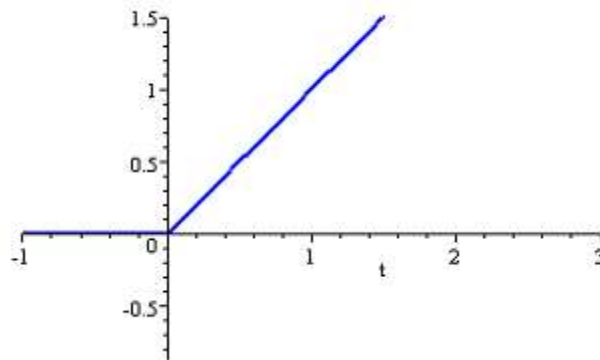


Figure 117

Example 89: Given that, $x(t) = -u(t + 3) + 2u(t + 1) - 2u(t - 1) + u(t - 3)$

Answer:

01. $-u(t + 3) \Rightarrow$

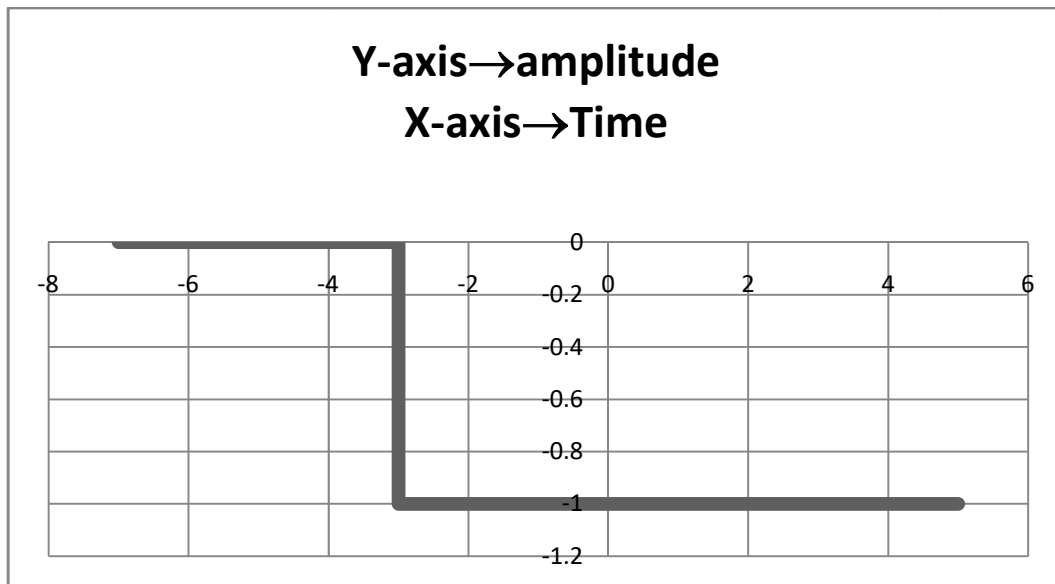
So,

$$-u(t + 3) = -1 ; t \geq -3;$$

$$\text{here, } t + 3 = 0$$

$$= 0; t < -3$$

$$\therefore t = -3$$



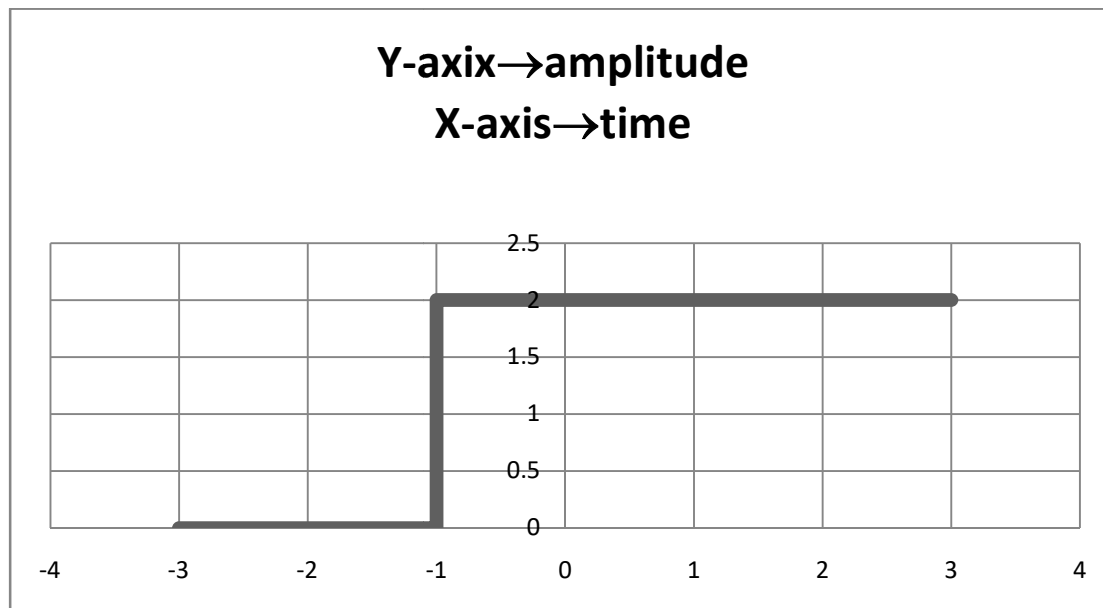
02. $2u(t + 1)$

$$2u(t + 1) = 2; t \geq -1$$

$$\text{here, } t + 1 = 0$$

$$= 0; t < -1$$

$$t = -1$$



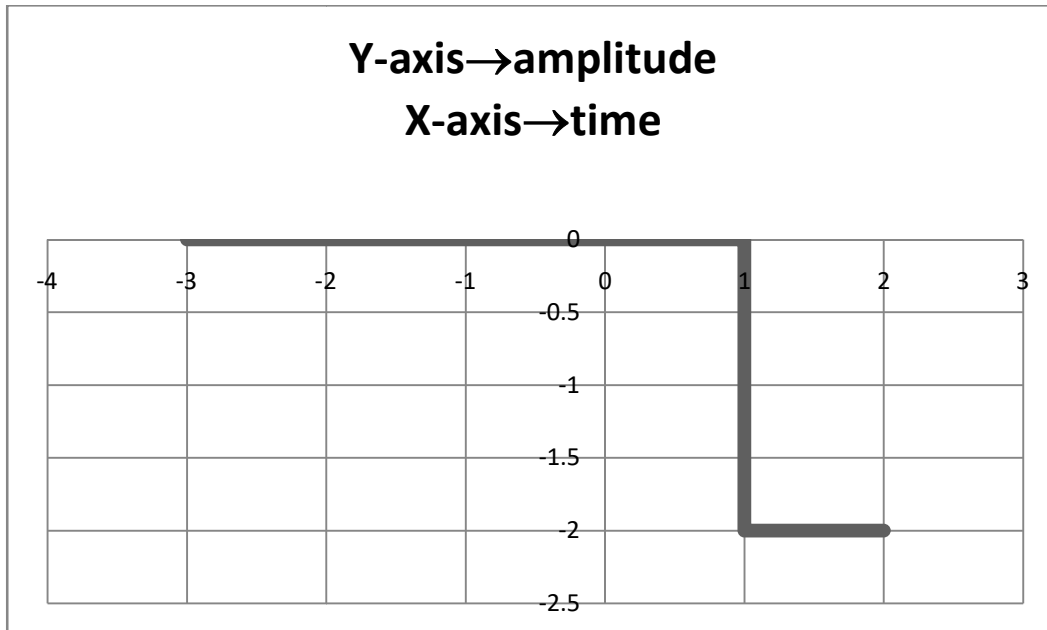
03. $-2u(t - 1)$

$$\therefore -2u(t - 1) = -2; t \geq 1$$

$$\text{here, } t - 1 = 0$$

$$= 0; t < 1$$

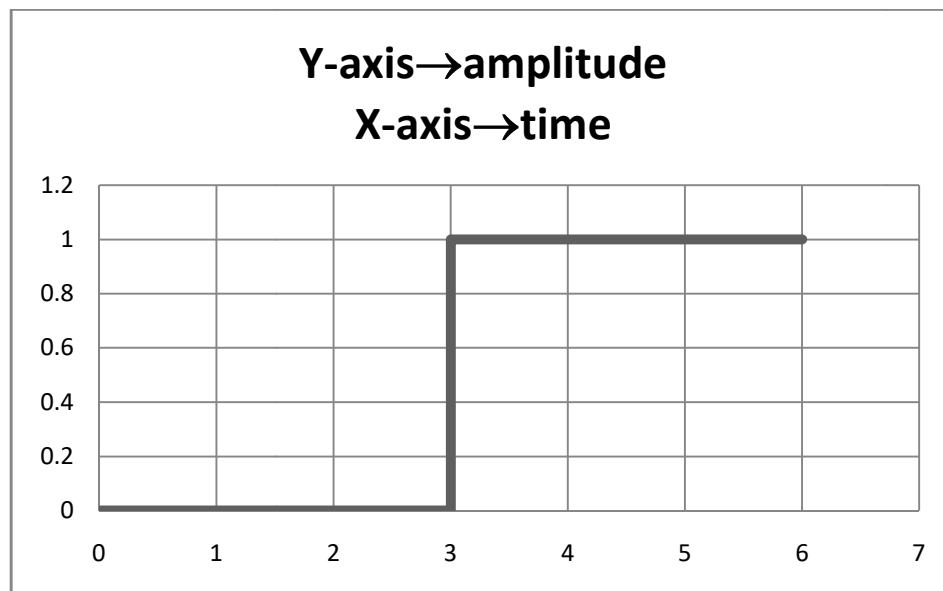
$$t = 1$$



04. $u(t - 3) \Rightarrow$

here, $t - 3 = 0$

$t = 3$

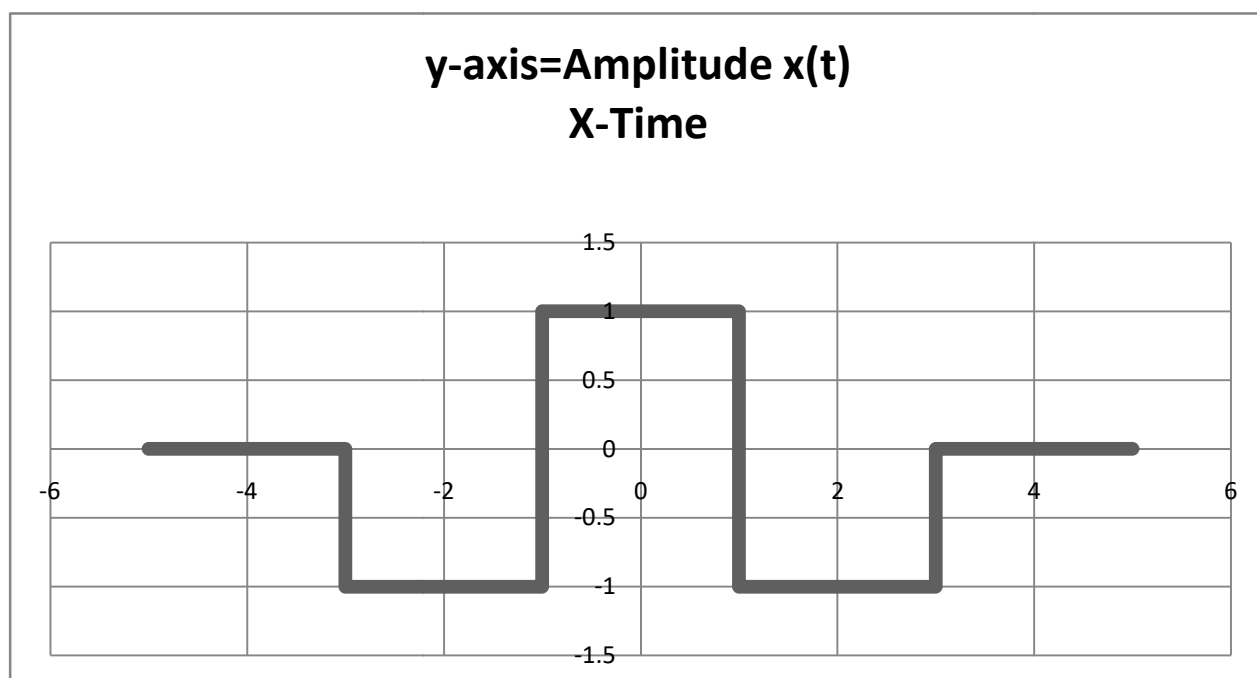


$$x(t) = -u(t + 3) + 2u(t + 1) - 2u(t - 1) + u(t - 3)$$

<https://www.youtube.com/watch?v=K2l3-Eb1YJ8>

<https://www.youtube.com/watch?v=Ee4XcAntNWQ>

<https://www.youtube.com/watch?v=r6pTvvT4lU>



Example 90:

$$x(t) = r(t + 2) - r(t + 1) - r(t - 1) + r(t - 2)$$

Solve:

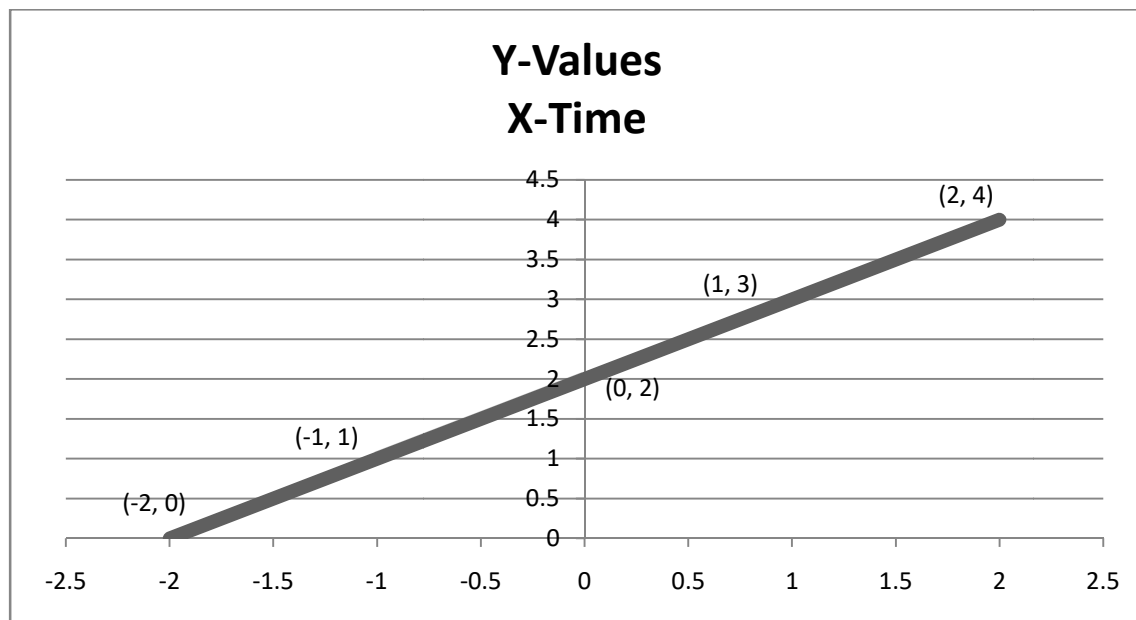
$$r(t + 2) = t + 2; t \geq -2$$

$$= 0; t < -2$$

$$\text{Here, } t + 2 = 0$$

$$\therefore t = -2$$

t	-2	-1	0	1	2	3
$r(t + 2) = t + 2$	0	1	2	3	4	5



Again,

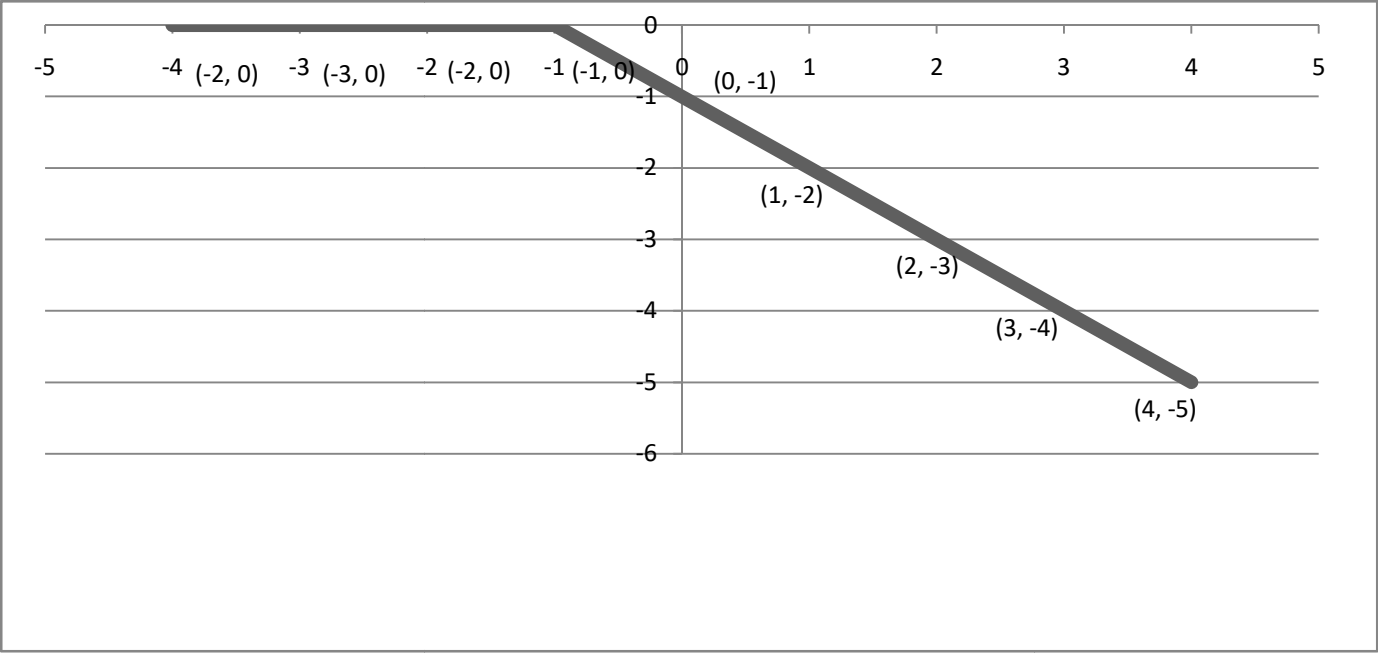
$$-r(t+1) = -(t+1); t \geq -1$$

$$= 0; t < -1$$

Here, $t+1 = 0$

$$\therefore t = -1$$

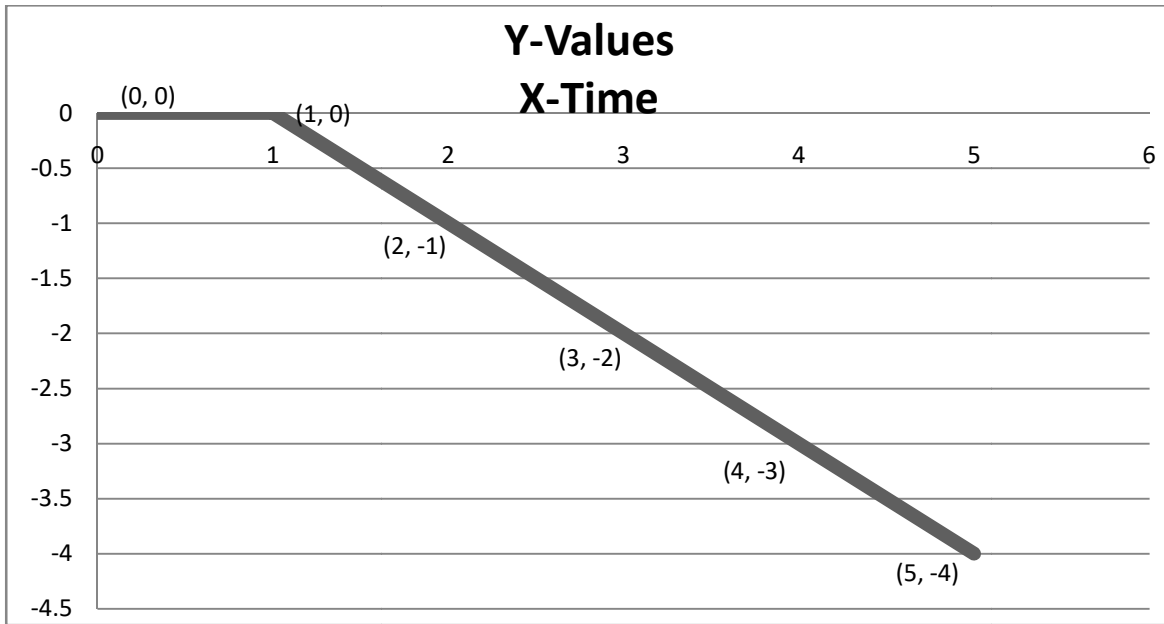
t	-1	0	1	2	3	4
$-r(t+1) = -(t+1)$	0	-1	-2	-3	-4	-5



$$\begin{aligned} \therefore -r(t-1) &= -(t-1) && ; t \geq 1 \\ &= 0 && ; t < 1 \end{aligned}$$

$$\begin{aligned} \text{Here, } t-1 &= 0 \\ \therefore t &= 1 \end{aligned}$$

t	1	2	3	4	5
$-r(t-1) = -(t-1)$	0	-1	-2	-3	-4

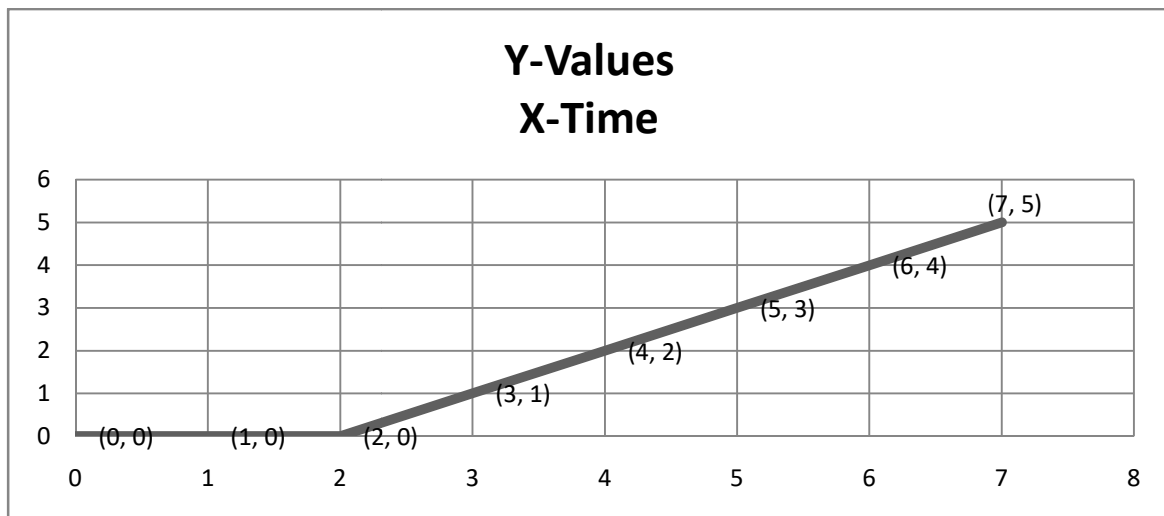


$$r(t - 2) = (t - 2) \quad ; t \geq 2$$

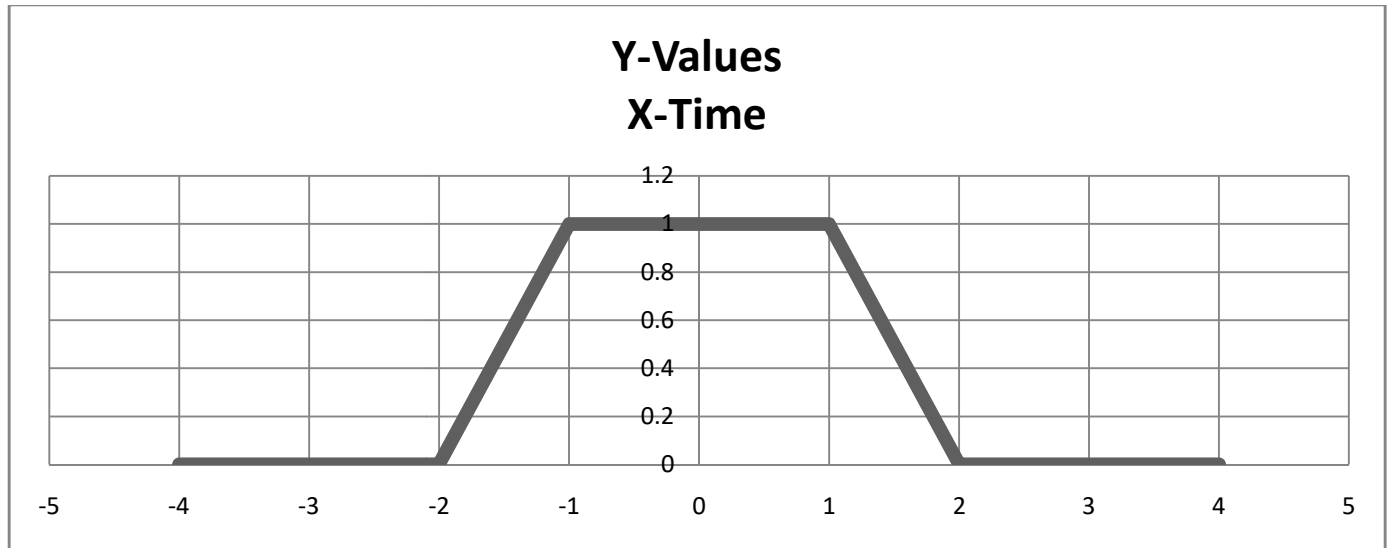
$$= 0 \quad ; t < 2$$

Here, $t - 2 = 0$
 $\therefore t = 2$

t	2	3	4	5	6	7
$r(t - 2) = t - 2$	0	1	2	3	4	5



$$\therefore x(t) = r(t + 2) - r(t + 1) - r(t - 1) + r(t - 2)$$



Problem 31: Impulse Function

The Impulse function $\delta(t)$ is defined as

$$\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{Otherwise} \end{cases}$$

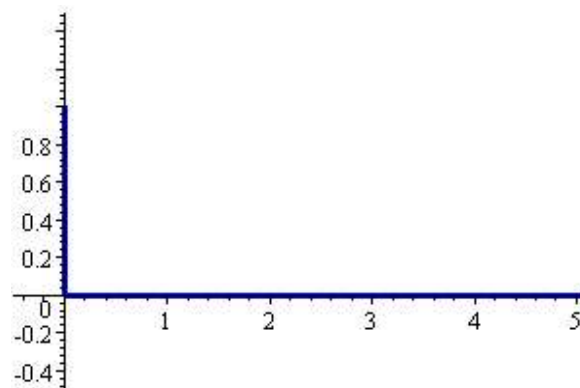
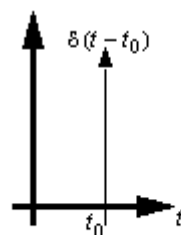


Figure 118



Delta (Impulse) Function

Figure 119

Example 91: Find $L[u(t-a)] = \frac{e^{-as}}{s}$

Proof: We have

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt$$

Here, $f(t) = u(t-a)$

$$L(f(t)) = \int_0^{\infty} f(t) e^{-st} dt$$

$$L(u(t-a)) = \int_0^{\infty} u(t-a) e^{-st} dt$$

$$L(u(t-a)) = \int_0^a 0 \cdot e^{-st} dt + \int_a^{\infty} 1 \cdot e^{-st} dt$$

[The unit function $u(t-a)$ is defined as

$$\begin{aligned} u(t) &= 1 & t &\geq a \\ &= 0 & t < a \end{aligned}$$

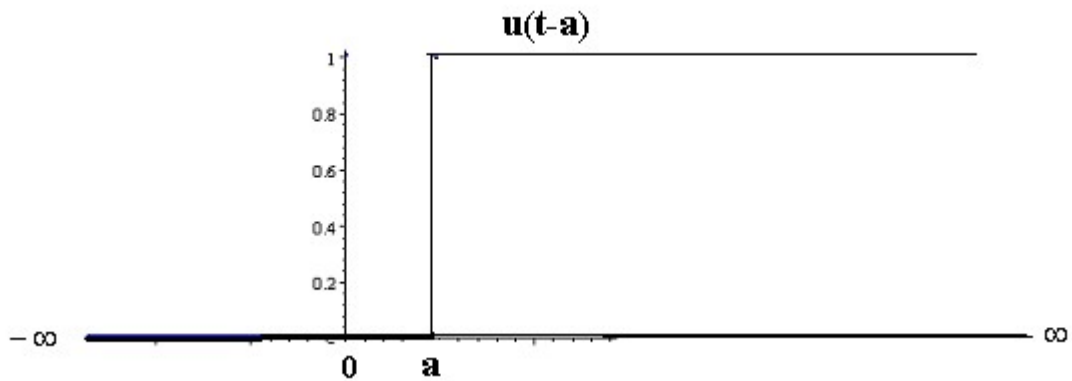


Figure 120

$$L(u(t-a)) = \int_0^a 0 \cdot e^{-st} dt + \int_a^{\infty} 1 \cdot e^{-st} dt$$

$$= 0 + \int_a^{\infty} 1 \cdot e^{-st} dt$$

$$= \int_a^{\infty} e^{-st} dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_a^{\infty}$$

$$= -\frac{1}{s} [e^{-s \times \infty} - e^{-s \times a}]$$

$$= -\frac{1}{s} [e^{-\infty} - e^{-as}]$$

$$\begin{aligned}
&= -\frac{1}{s} \left[\frac{1}{e^\infty} - e^{-as} \right] \\
&= -\frac{1}{s} \left[\frac{1}{\infty} - e^{-as} \right] \\
&= -\frac{1}{s} [0 - e^{-as}] \\
&= +\frac{1}{s} [e^{-as}] \\
&= \frac{e^{-as}}{s}
\end{aligned}$$

$$\therefore L(u(t-a)) = \frac{e^{-as}}{s} \text{ (Proved)}$$

$$\therefore L(u(t-2)) = \frac{e^{-2s}}{s}$$

Example 92: Express the following function in terms of unit step functions and find its Laplace transform:

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

Solution:

We have

$$L(u(t-a)) = \frac{e^{-as}}{s}$$

$$\therefore L(u(t-2)) = \frac{e^{-2s}}{s} \text{----- (i)}$$

Given,

$$f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$$

$$f(t) = \begin{cases} 8+0; & t < 2 \\ 8-2; & t > 2 \end{cases}$$

$$f(t) = 8 + \begin{cases} 0; & t < 2 \\ -2; & t > 2 \end{cases}$$

$$f(t) = 8 + (-2) \begin{cases} 0; & t < 2 \\ 1; & t > 2 \end{cases}$$

$$f(t) = 8 + (-2) \begin{cases} 1; & t > 2 \\ 0; & t < 2 \end{cases}$$

$$f(t) = 8 + (-2)u(t - 2)$$

$$f(t) = 8 - 2u(t - 2)$$

$$L\{f(t)\} = L\{8 - 2u(t - 2)\}$$

$$L\{f(t)\} = L\{8\} - 2L\{u(t - 2)\}$$

$$L\{f(t)\} = 8L\{1\} - 2L\{u(t - 2)\}$$

$$L\{f(t)\} = 8 \times \frac{1}{s} - 2 \times \frac{e^{-2s}}{s}$$

$$[\because L(1) = \frac{1}{s} \text{ (from example 55)}] \text{ \& } L[u(t - 2)] = \frac{e^{-2s}}{s} \text{ (from example 91)]}$$

Problem 32:

Product of $u(t)$ vs. Shifting the Function Along the t -axis

1. $f(t) \cdot u(t)$

The $f(t)$ part begins at $t = 0$

2. $f(t) \cdot u(t - a)$

The $f(t)$ part begins at $t = a$

3. $f(t - a) \cdot u(t)$

The $f(t)$ part has been **shifted** to the right by a units and begins at $t = 0$

4. $f(t - a) \cdot u(t - a)$

The $f(t)$ part has been **shifted** to the right by a units and begins at $t = a$

Example 93: $g(t) = \sin t \cdot u(t)$

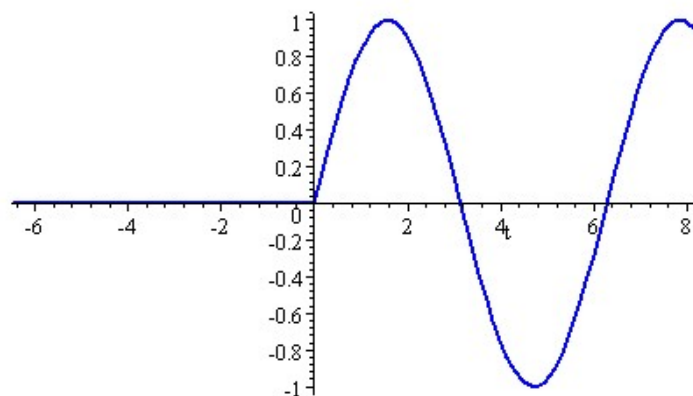


Figure 121: the $\sin t$ part starts at $t = 0$

Example 94: $g(t) = \sin t \cdot u(t - 0.7)$

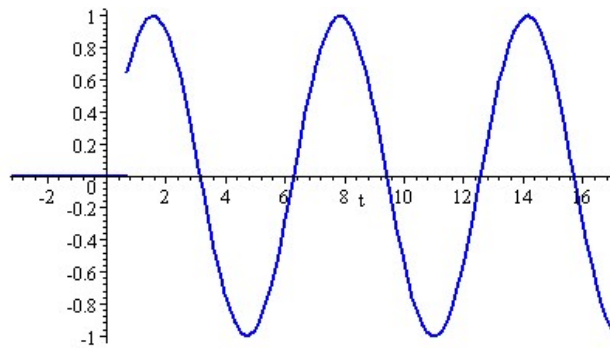


Figure 122: the $\sin t$ part starts at $t = 0.7$

Example 95: $g(t) = \sin(t - 0.7) \cdot u(t)$

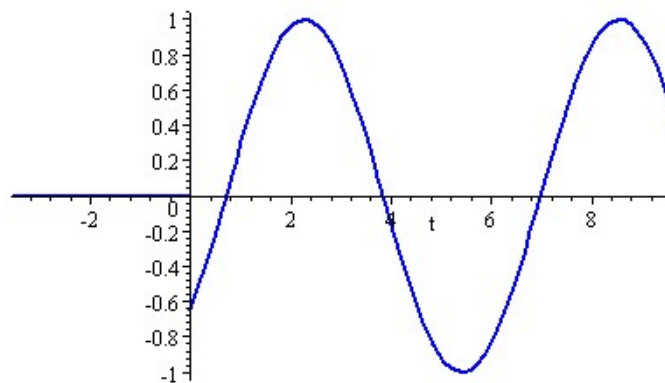


Figure 123: the $\sin t$ part has been shifted 0.7 units to the right, and it starts at $t = 0$

Example 96: $g(t) = \sin(t - 0.7) \cdot u(t - 0.7)$

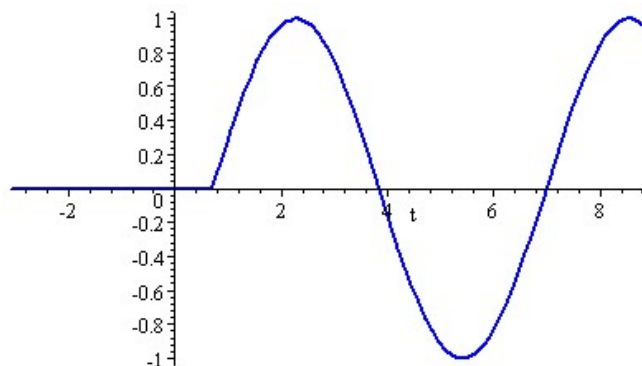


Figure 124: the $\sin t$ part has been shifted 0.7 units to the right, and it starts at $t = 0.7$

Example 97: If $f(t) = \sin t$ then the graph of $g(t) = \sin t \cdot u(t - 2\pi)$ is

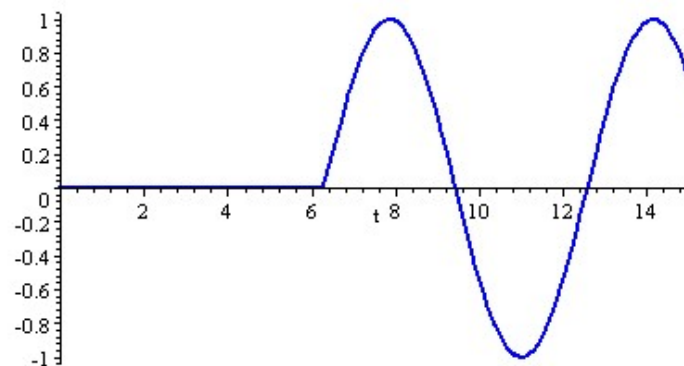


Figure 125: The $\sin t$ portion starts at $t = 2\pi$ because we have multiplied $\sin t$ by $u(t - 2\pi)$

Example 98: If $f(t) = 10e^{-2t}$, then the graph of $g(t) = 10e^{-2t} \cdot u(t - 5)$ is

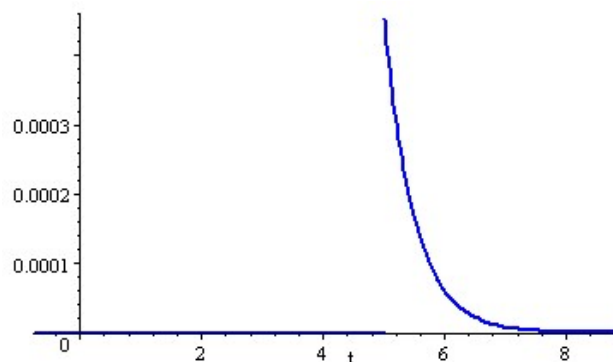


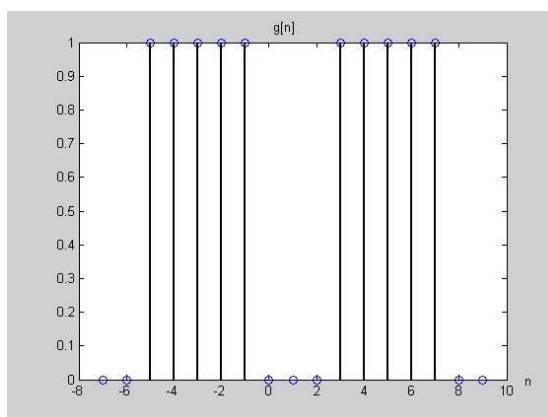
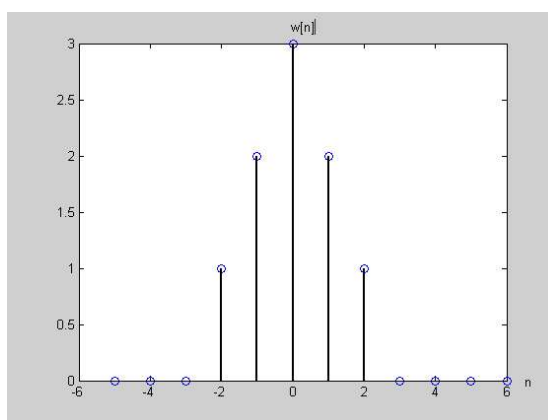
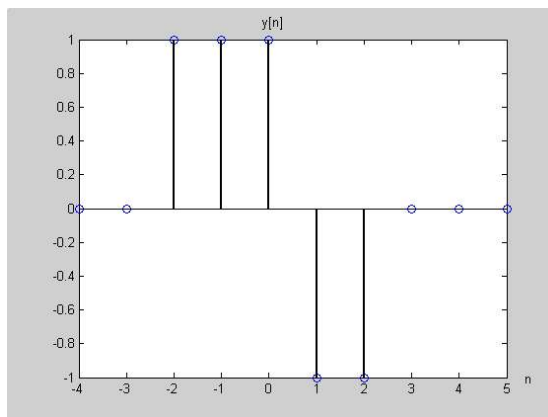
Figure 126: The portion $10e^{-2t}$ starts at $t = 5$

Home Task:

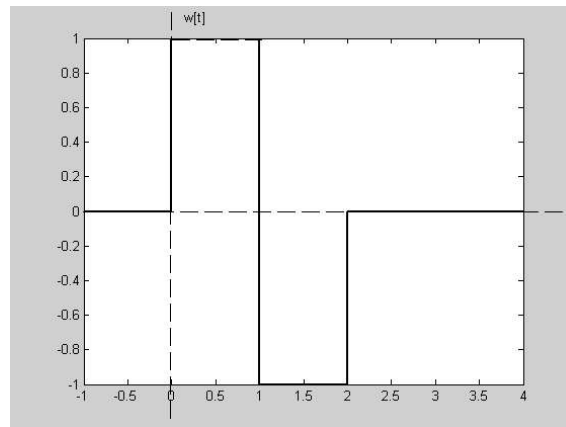
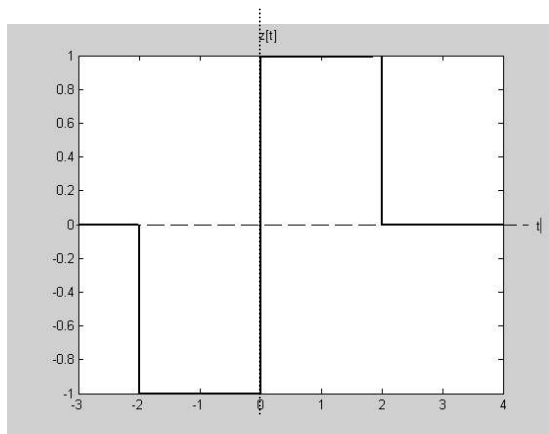
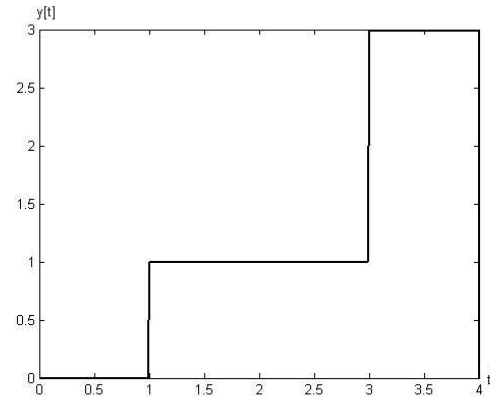
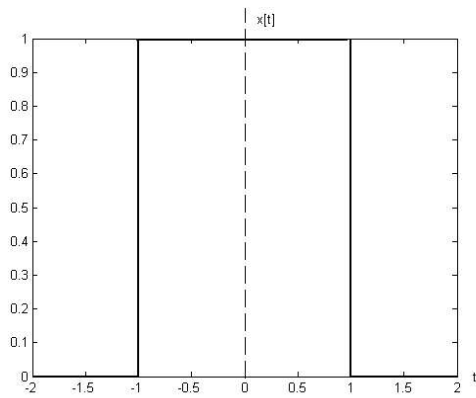
Draw the graph of $f(t) = 4u(t) - 8u(t - 1) + 4u(t - 2)$

1. Consider the discrete-time signals depicted in the following figures. Evaluate the convolution sums indicated below:

1. $m[n] = x[n] * z[n]$
2. $m[n] = x[n] * y[n]$
3. $m[n] = x[n] * f[n]$
4. $m[n] = x[n] * g[n]$
5. $m[n] = y[n] * z[n]$
6. $m[n] = y[n] * g[n]$
7. $m[n] = y[n] * w[n]$
8. $m[n] = y[n] * f[n]$
9. $m[n] = z[n] * g[n]$
10. $m[n] = w[n] * g[n]$
11. $m[n] = f[n] * g[n]$



- i) $m(t) = x(t) * y(t)$
- ii) $m(t) = x(t) * z(t)$
- iii) $m(t) = y(t) * z(t)$
- iv) $m(t) = y(t) * w(t)$



3. Sketch the waveforms of the following signals:

a) $x(t) = u(t) - u(t-2)$

b) $x(t) = u(t+1) - 2u(t) + u(t-1)$

c) $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

d) $x(t) = 4u(t-1) - 8u(t-4) + 4u(t-6)$

e) $x(t) = r(t-1) + u(t) - u(t-1) - 2r(t-2) + r(t-3)$

f) $x(t) = r(t+1) - r(t) + r(t-2)$

g) $x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$

h) $x(t) = r(t+2) - 2r(t+1) + r(t-1)$

Problem 33: Inverse Laplace transform

Example 99:

If $L[f(t)] = F(S)$, Then $L^{-1}[F(S)] = f(t)$

Where L^{-1} is called the inverse Laplace transform operator.

$$(1) L^{-1}\left(\frac{1}{s}\right) = 1 \left[\because (1) L[f(t)] = L(1) = \frac{1}{s} \right]$$

$$(2) L^{-1}\left(\frac{a}{s}\right) = a \left[\because L(f(t)) = L(a) = \frac{a}{s} \right]$$

$$(3) L^{-1}\left(\frac{1}{s^2}\right) = t \left[\because L(f(t)) = L(t) = \frac{1}{s^2} \right]$$

$$(4) L^{-1}\left(\frac{1}{s-a}\right) = e^{at} \left[\because L(f(t)) = L(e^{at}) = \frac{1}{s-a} \right]$$

$$(5) L^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \left[\because L(f(t)) = L(e^{2t}) = \frac{1}{s-2} \right]$$

$$(6) L^{-1}\left(\frac{a}{s^2 + a^2}\right) = \sin at \left[\because L(f(t)) = L(\sin at) = \frac{a}{s^2 + a^2} \right]$$

$$(7) L^{-1}\left(\frac{2}{s^2 + 2^2}\right) = \sin 2t \left[\because L(f(t)) = L(\sin 2t) = \frac{2}{s^2 + 2^2} \right]$$

$$(8) L^{-1}\left(\frac{s}{s^2 + a^2}\right) = \cos at \left[\because L(f(t)) = L(\cos at) = \frac{s}{s^2 + a^2} \right]$$

$$(9) L^{-1}\left(\frac{s}{s^2 + 2^2}\right) = \cos 2t \left[\because L(f(t)) = L(\cos 2t) = \frac{s}{s^2 + 2^2} \right]$$

$$(10) L^{-1}\left(\frac{a}{s^2 - a^2}\right) = \sin hat \left[\because L(f(t)) = L(\sin hat) = \frac{a}{s^2 - a^2} \right]$$

$$(11) L^{-1}\left(\frac{2}{s^2 - 2^2}\right) = \sin h2t \left[\because L(f(t)) = L(\sin h2t) = \frac{2}{s^2 - 2^2} \right]$$

$$(12) L^{-1}\left(\frac{s}{s^2 - a^2}\right) = \cos hat \left[\because L(f(t)) = L(\cos hat) = \frac{s}{s^2 - a^2} \right]$$

$$(13) L^{-1}\left(\frac{s}{s^2 - 2^2}\right) = \cos h2t \left[\because L(f(t)) = L(\cos h2t) = \frac{s}{s^2 - 2^2} \right]$$

Example 100 Find the inverse Laplace transforms of the following:

$$(i) \frac{1}{s-2}, (ii) \frac{s}{s^2-16}, (iii) \frac{5}{s^2+25}, (iv) \frac{1}{(s+3)^2-4}, (v) \frac{1}{2s-7}$$

Answers:

$$(i) \text{ Given } f(s) = \frac{1}{s-2}$$

$$\text{We have, } \therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a}$$

[Example 58]

$$\therefore e^{at} = L^{-1}\left(\frac{1}{s-a}\right)$$

$$\therefore e^{2t} = L^{-1}\left(\frac{1}{s-2}\right)$$

$$\therefore L^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \text{ Answer}$$

$$(ii) \quad L^{-1}\left(\frac{s}{s^2 - 16}\right) = ?$$

$$\text{We have, } \therefore L(f(t)) = L(\cosh at) = \frac{s}{s^2 - a^2}$$

[Example 63]

$$\therefore \cosh at = L^{-1}\left(\frac{s}{s^2 - a^2}\right)$$

$$\therefore \cosh 4t = L^{-1}\left(\frac{s}{s^2 - 4^2}\right)$$

$$\therefore L^{-1}\left(\frac{s}{s^2 - 4^2}\right) = \cosh 4t \quad \text{Answer}$$

$$(iii) \text{ We have, } \therefore L(f(t)) = L(\sin at) = \frac{a}{s^2 + a^2}$$

[Example 60]

$$\therefore \sin at = L^{-1}\left(\frac{a}{s^2 + a^2}\right)$$

$$\therefore \sin 5t = L^{-1}\left(\frac{5}{s^2 + 5^2}\right)$$

$$\therefore L^{-1}\left(\frac{5}{s^2 + 5^2}\right) = \sin 5t$$

$$\therefore L^{-1}\left(\frac{5}{s^2 + 5^2}\right) = \sin 5t \quad \text{Answer}$$

Example 101:

Find inverse Laplace transform of: $\frac{s+4}{s(s-1)(s-2)}$

Solution:

Let,

$$\frac{s+4}{s(s-1)(s-2)} = \frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s-2)} \dots\dots\dots (i)$$

Multiplying by $s(s-1)(s-2)$ in both sides

$$\Rightarrow \frac{s+4}{s(s-1)(s-2)} \times s(s-1)(s-2) = A \frac{s(s-1)(s-2)}{s} + B \frac{s(s-1)(s-2)}{(s-1)} + C \frac{s(s-1)(s-2)}{(s-2)}$$

$$\Rightarrow s+4 = A(s-1)(s-2) + Bs(s-2) + Cs(s-1) \dots\dots\dots (ii)$$

Put $s = 0$ in equation (ii),

$$\Rightarrow 0+4 = A(0-1)(0-2) + B \times 0(0-2) + C \times 0(0-1)$$

$$\Rightarrow 4 = 2A$$

$$\therefore A = 2$$

Put $s - 1 = 0$, i.e. $s = 1$ in equation (ii),

$$\Rightarrow 1 + 4 = A(1 - 1)(1 - 2) + B \times 1(1 - 2) + C \times 1(1 - 1)$$

$$\Rightarrow 5 = 0 - B + 0$$

$$\therefore B = -5$$

Put $s - 2 = 0$, i.e. $s = 2$ in equation (ii),

$$\Rightarrow 2 + 4 = A(2 - 1)(2 - 2) + B \times 2(2 - 2) + C \times 2(2 - 1)$$

$$\Rightarrow 6 = 0 + 0 + C(4 - 2)$$

$$\Rightarrow 6 = 0 + 0 + 2C$$

$$\therefore C = 3$$

Putting the value of A, B, C in equation (i), we get,

$$\frac{s + 4}{s(s - 1)(s - 2)} = \frac{A}{s} + \frac{B}{(s - 1)} + \frac{C}{(s - 2)}$$

$$\frac{s + 4}{s(s - 1)(s - 2)} = \frac{2}{s} + \frac{-5}{(s - 1)} + \frac{3}{(s - 2)}$$

$$\therefore L^{-1}\left(\frac{s + 4}{s(s - 1)(s - 2)}\right) = L^{-1}\left(\frac{2}{s}\right) + L^{-1}\left(\frac{-5}{s - 1}\right) + L^{-1}\left(\frac{3}{s - 2}\right)$$

$$L^{-1}\left(\frac{s + 4}{s(s - 1)(s - 2)}\right) = 2L^{-1}\left(\frac{1}{s}\right) - 5L^{-1}\left(\frac{1}{s - 1}\right) + 3L^{-1}\left(\frac{1}{s - 2}\right) \text{-----}(iii)$$

Since

$$01. \text{ We have } \therefore L(f(t)) = L(1) = \frac{1}{s} \quad [\text{Example 55}]$$

$$\therefore 1 = L^{-1}\left(\frac{1}{s}\right)$$

$$\therefore L^{-1}\left(\frac{1}{s}\right) = 1$$

$$02. \text{ We have, } \therefore L(f(t)) = L(e^{at}) = \frac{1}{s - a} \quad [\text{Example 58}]$$

$$\therefore e^{at} = L^{-1}\left(\frac{1}{s - a}\right)$$

$$\therefore e^t = L^{-1}\left(\frac{1}{s - 1}\right)$$

$$\therefore L^{-1}\left(\frac{1}{s - 1}\right) = e^t$$

$$03. \therefore L(f(t)) = L(e^{at}) = \frac{1}{s-a} \quad [\text{Example 58}]$$

$$\therefore e^{at} = L^{-1}\left(\frac{1}{s-a}\right)$$

$$\therefore e^{2t} = L^{-1}\left(\frac{1}{s-2}\right)$$

$$\therefore L^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \text{ Answer}$$

Putting these values in (iii), we get

$$\begin{aligned} L^{-1}\left(\frac{s+4}{s(s-1)(s-2)}\right) &= 2L^{-1}\left(\frac{1}{s}\right) - 5L^{-1}\left(\frac{1}{s-1}\right) + 3L^{-1}\left(\frac{1}{s-2}\right) \\ &= 2.1 - 5e^t + 3e^{2t} \text{ Answer} \end{aligned}$$

Problem 34: Convolution Sum

Convolution is a mathematical tool for combining two signals to produce a third signal.

The following steps are to be taken

- i. Folding
- ii. Shifting
- iii. Multiplication
- iv. Summation

1st times:

- i. Folding
- ii. Multiplication
- iii. Summation

2nd times and more

- i. Shifting
- ii. Multiplication
- iii. Summation

Example 102:

Evaluate the convolution sums of $y[n] = x[n] * h[n]$

Where,

$x[n] = 1, n = 0$ and $h[n] = 2, n = 0$; n represents the time index
 $3, n = 1$ $1, n = 1$

MATLAB

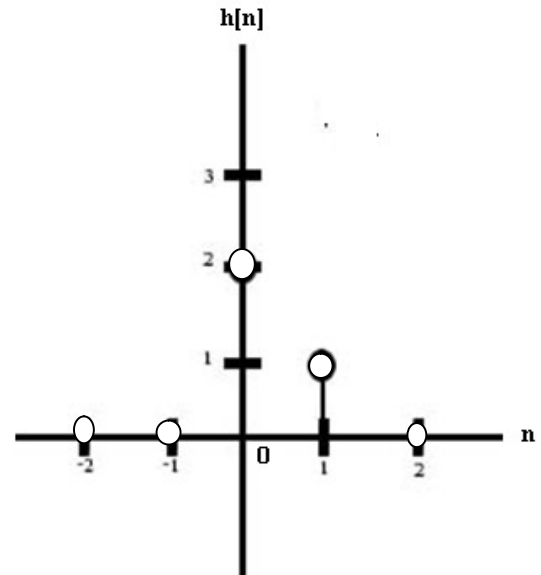
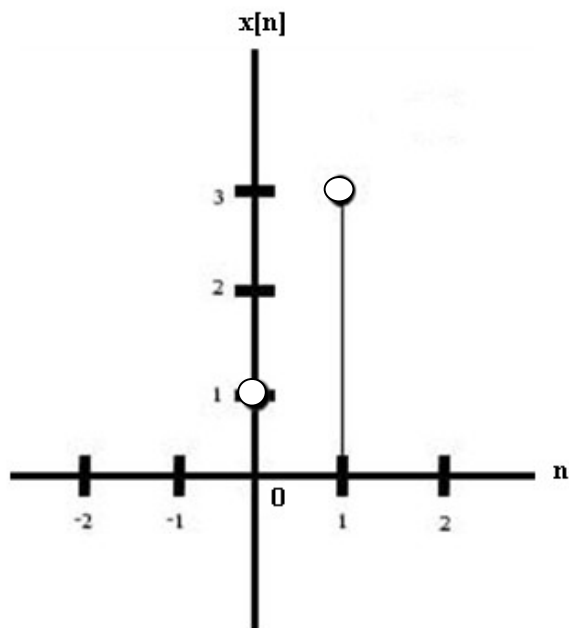
$x=[1 \ 3]$

$h=[2 \ 1]$

$y=\text{conv}(x,h)$

$y= [2 \ 7 \ 3]$

Solution:



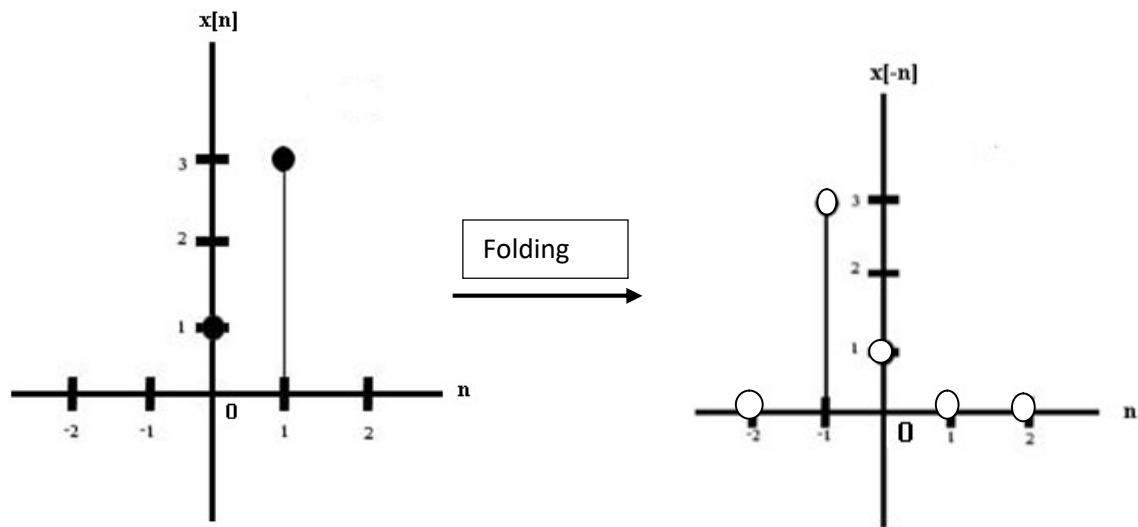
1st time:

(i). Folding:

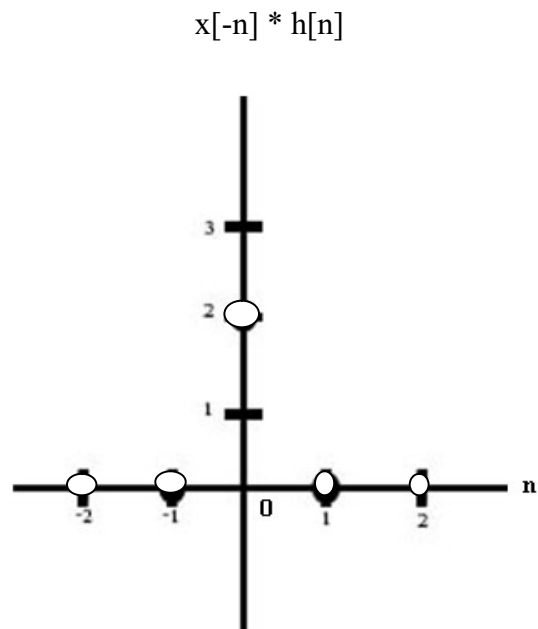
$$x[n] = x[-n]$$

$$\text{i.e. } x[0] = x[0]$$

$$x[1] = x[-1]$$



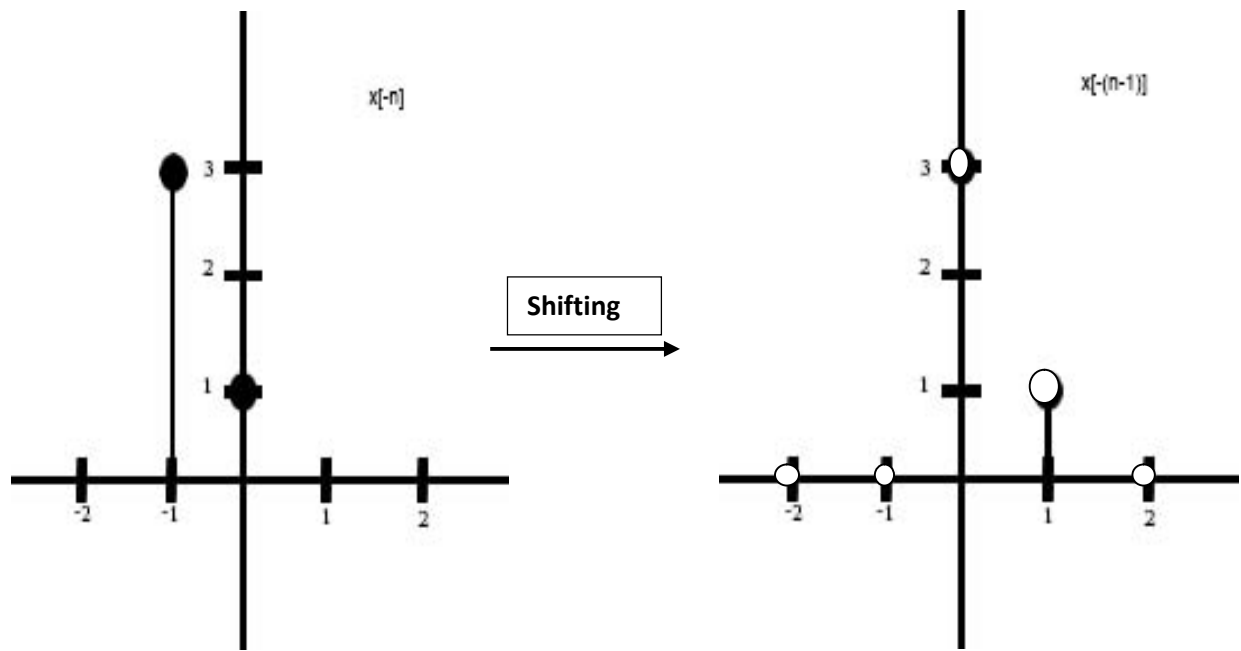
(ii). Multiplication:



(iii). Summation: $y[0] = 0 + 2 + 0 = 2$

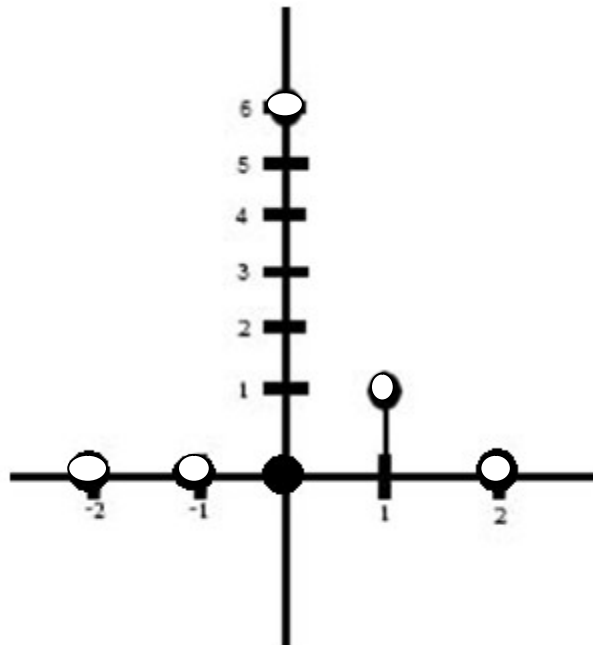
2nd time:

(i). Shifting:



(ii) Multiplication:

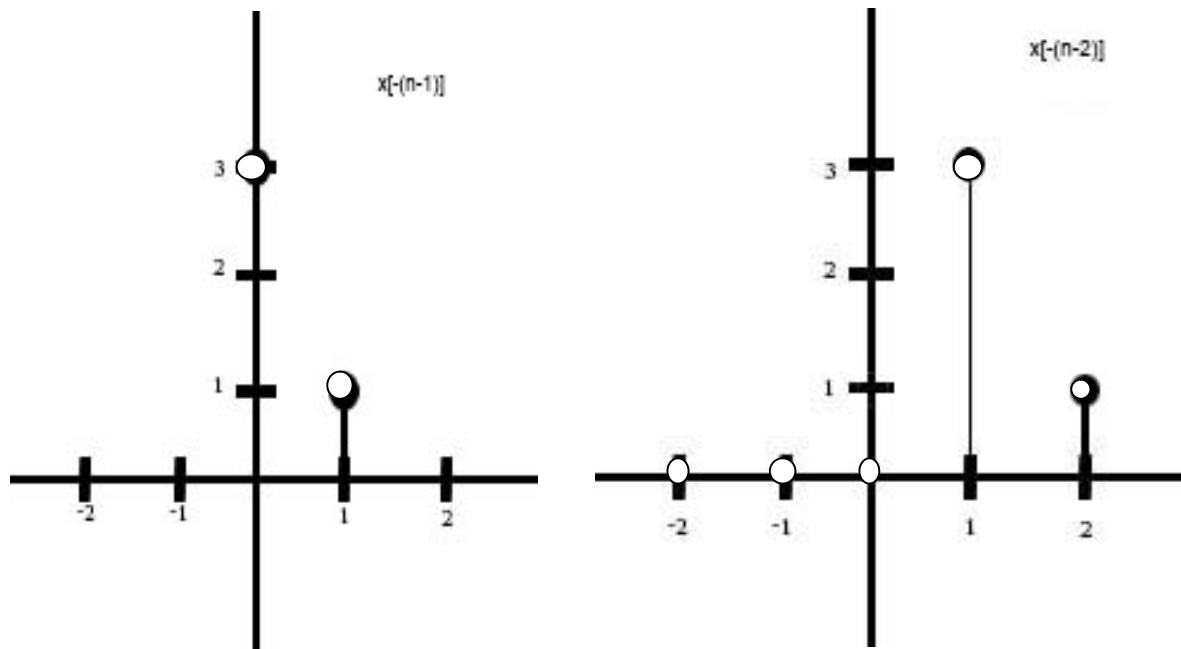
$$x[-(n-1)] * h[n]$$



(iii). Summation: $y[1] = 0 + 6 + 1 + 0 = 7$

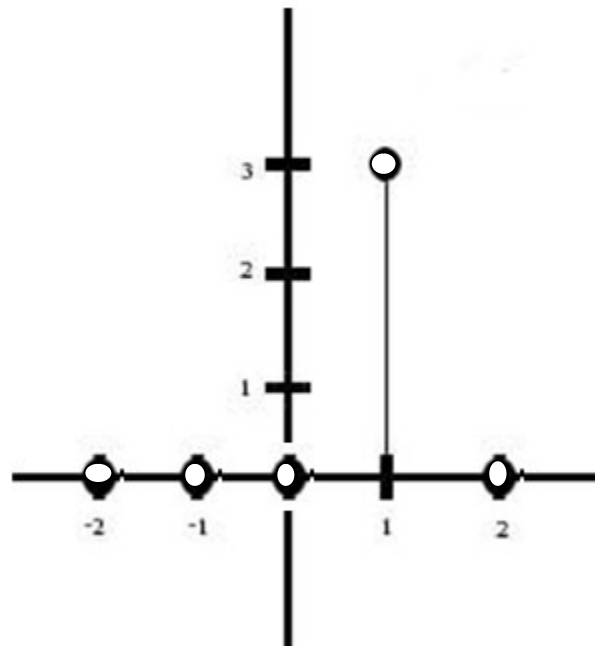
3rd time:

(i). Shifting:



(ii). Multiplication:

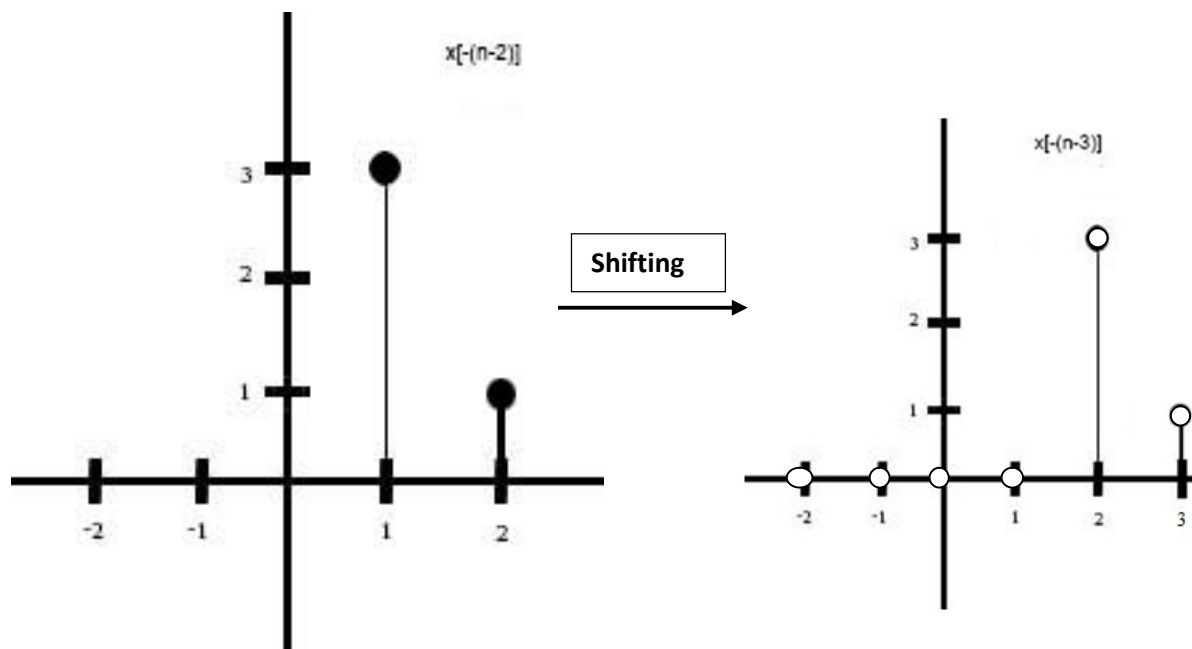
$$x[-(n-2)] * h[n]$$



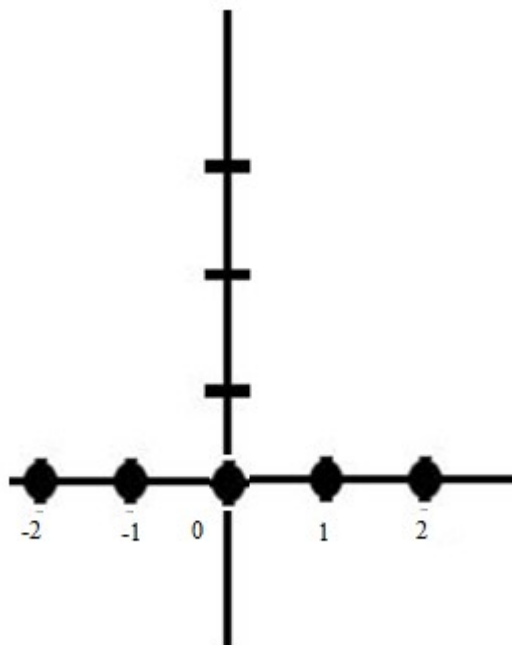
(iii). Summation: $y[2] = 0 + 0 + 3 + 0 = 3$

4th time:

(i). Shifting:



(ii). Multiplication: $x[-(n-3)] * h[n]$



iii) Summation: $y[3] = 0+0+0+0=0$

Finally we get,

$$\therefore y[0] = 2$$

$$y[1] = 7$$

$$y[2] = 3$$

$\therefore$ Convolution Sum: $y[n] = [2 \ 7 \ 3]$

